

Feasibility governor for MPC with disturbance preview information

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ABSTRACT

This paper presents a feasibility governed preview MPC (PMPC) strategy for the control of constrained linear systems with previewed disturbances. The feasibility governor (FG) minimally modifies the supplied reference command so that the predicted terminal state of the PMPC policy remains in a robustly control invariant terminal set. As a result, recursively feasibility of the combined FG-PMPC strategy can be achieved with shorter prediction horizons than a standard PMPC strategy. Hence, the amount of computational effort and preview information necessary to implement PMPC is reduced. Conditions are established that guarantee finite-time convergence of the FG reference command and input-to-state stability of the closed-loop system. An adaptive cruise control example is reported to demonstrate a scenario where both preview information and the FG are necessary to ensure safety and real-time feasibility of the PMPC policy.

1. Introduction

Model predictive control (MPC) is a well-known feedback strategy defined by the solution of a receding horizon optimal control problem (OCP). The stability and robustness of MPC have been studied extensively (e.g., [1–3]) and various methods have been developed to handle cases where disturbances are present in the system. If the disturbances are sufficiently small, then the inherent robustness of nominal MPC may be enough to guarantee robust stability [4–6]. If the disturbances are set-bounded, then robust MPC strategies (e.g., tube-based [7] and min-max [8]) can provide guarantees of recursive feasibility and robust stability. Finally, if the disturbances are stochastic, then stochastic MPC strategies can provide probabilistic guarantees of constraint satisfaction and feasibility [9]. For a more complete review of such methods, we refer the reader to the extensive literature on robust and stochastic MPC (e.g., [1,3,9,10]).

In this paper, we are interested in the particular case where a preview of the disturbance is available. This is motivated by practical applications where measurements, estimates, or forecasts of the disturbance are possible, for example: weather forecasting in renewable energy systems [11]; gust detection in wind turbines [12] and UAVs [13,14]; road terrain profiling in automotive suspension and braking systems [15–17]; and path tracking and electronic horizons in autonomous driving [18,19].¹ In many of these applications, the

so-called disturbances² may arise from changing environmental conditions that can be accurately measured/predicted (e.g., road terrain and curvature conditions [15–19]). In such scenarios, the disturbances can be accounted for by including this preview information in the OCP that defines the MPC law. Then, the resulting preview MPC (PMPC) law can either exploit or mitigate the effect of future disturbances — depending on whether the disturbances are acting favorably or unfavorably. Hence, if an accurate preview is available, then PMPC will provide superior performance to robust MPC strategies that assume worst-case realizations of the disturbance.

The efficacy of PMPC has been demonstrated in practical applications (e.g., [12–18]), however, there are still relatively few papers that explore the stability characteristics of these controllers in a general setting. In a recent paper, Fang and Chen [20] developed an input-to-state stability (ISS) framework to analyze nonlinear systems controlled by PMPC with exact preview information over the prediction horizon. Monasterios and Trodden [21] showed that robust stability and recursive feasibility can be guaranteed for linear systems controlled by PMPC if the preview has bounded error, the current disturbance is known exactly, and the disturbances are equal to a known constant at timesteps beyond the MPC prediction horizon.

Both [20] and [21] rely on the enforcement of a robustly control invariant (RCI) terminal constraint in the OCP to guarantee recursive feasibility. However, enforcement of such a terminal constraint may

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¹ For even more examples of such applications, we refer readers to [20,21,32], and any of the aforementioned references.

² Preview control strategies can also be used to address control problems with changing reference commands (e.g., see [32]). In these scenarios, the reference command changes are previewed and modeled in same way as a measured exogenous disturbance (e.g., see [32, Section 3]). Hence, we use the term “disturbance” to also include changing reference commands.

require the use of a long prediction horizon if the initial state is far from the target equilibrium, the system constraints are restrictive, and/or the disturbance is acting in an unfavorable manner. The necessity of a long prediction horizon poses two major problems. First, the computational cost of solving the OCP grows with the prediction horizon. This is also a concern for standard MPC laws, however, this issue is compounded in PMPC because the disturbance can make reachability of the terminal set more difficult. Moreover, the RCI terminal sets used in [20] and [21] are smaller than the control invariant terminal sets that can be used in standard MPC. Second, it may not be practically feasible to have accurate preview information over a long prediction horizon.

Hence, the prediction horizon of PMPC must be chosen to satisfy competing interests when standard approaches are adopted. That is, the prediction horizon must be: (1) short enough to adhere to limitations imposed by onboard computational resources; (2) no longer than the duration for which preview information is available; but also, (3) long enough to enforce reachability to an RCI terminal set. To address this conundrum, we propose a feasibility governed PMPC strategy in this paper. In particular, we extend the feasibility governor (FG) framework introduced in [22] and [23] so that it can be applied to the PMPC framework of [20]. The proposed FG strategy minimally modifies the reference command supplied to the PMPC problem so that the predicted terminal state is always contained in an RCI terminal region for the augmented reference command.

The theoretical contributions of our work are strong guarantees of input-to-state stability (ISS) of the resulting FG-PMPC closed-loop system and finite-time convergence of the FG reference command. These guarantees are derived by extending the ISS analysis framework of [20] to a reference tracking PMPC setting. Then, ISS of the closed-loop PMPC system with a constant reference is used to prove ISS and convergence of the resulting FG-PMPC system. The practical contribution of our work is a supervisory strategy that improves the real-time feasibility of PMPC by ensuring recursive feasibility with much shorter prediction horizons. As a result, the PMPC horizon length can be chosen to adhere to practical limitations (i.e., maximum computation time and maximum preview information), while still ensuring recursive feasibility and ISS of the closed-loop system.

The paper is organized as follows. Section 2 describes the reference tracking PMPC strategy used throughout. Section 3 extends the ISS results of [20] to reference tracking. Section 4 describes the proposed FG-PMPC strategy and establishes proofs of finite-time convergence of the FG reference command and ISS of the resulting closed-loop system. Section 5 showcases an adaptive cruise control example where both preview information and the FG are necessary to ensure safety and real-time feasibility of the PMPC policy. Finally, Section 6 contains concluding remarks.

Notation: Let $\mathbb{N}$ represent the natural numbers including 0. Given $a, b \geq 0$, let $\mathbb{N}_{[a,b]} = \mathbb{N} \cap [a, b]$. Given $x \in \mathbb{R}^n$ and $y \in \mathbb{R}^m$, let $(x, y) = [x^T \ y^T]^T$. Given a set $\Gamma \subseteq \mathbb{R}^{n+m}$ and a parameter $x \in \mathbb{R}^n$, let $S_y(\Gamma, x) = \{y \in \mathbb{R}^m \mid (x, y) \in \Gamma\}$. Given $W \in \mathbb{R}^{n \times n}$, $W > 0$ means that W is symmetric positive-definite (≥ 0 , semi-definite), and the W -norm of $x \in \mathbb{R}^n$ is $\|x\|_W = \sqrt{x^T W x}$. Let $\|\cdot\|$ represent the 2-norm when no subscript is specified. Given $\rho \in \mathbb{R}$ and $X \subset \mathbb{R}^n$, let $\rho X = \{\rho x \mid x \in X\}$. For a sequence $\{v_i\}_{i=0}^\infty$, the notation $v_i \equiv v$ is used as shorthand for $v_i = v$ for all $i \in \mathbb{N}$. The classes of $\mathcal{K}$, $\mathcal{K}_\infty$, and $\mathcal{KL}$ functions follow the usual definitions in nonlinear systems theory (e.g., see [24]).

2. Problem setting

We consider linear time-invariant systems that can be represented by models of the form

$$x_{k+1} = Ax_k + Bu_k + w_k, \quad (1a)$$

$$y_k = Cx_k + Du_k, \quad (1b)$$

$$z_k = Ex_k + Fu_k, \quad (1c)$$

where $x_k \in \mathbb{R}^n$ is the state, $u_k \in \mathbb{R}^u$ is the control, $w_k \in \mathbb{R}^n$ is a known disturbance, $y_k \in \mathbb{R}^y$ is the constrained output, and $z_k \in \mathbb{R}^z$ is the tracking output. The disturbance is assumed to belong to a known set:

$$w_k \in \mathcal{W}, \quad \forall k \in \mathbb{N}. \quad (2)$$

The control objective is to asymptotically track a desired reference $r \in \mathbb{R}^z$ with the tracking output z_k subject to constraints

$$y_k \in \mathcal{Y}, \quad \forall k \in \mathbb{N}, \quad (3)$$

where $\mathcal{Y} \subseteq \mathbb{R}^y$ is a specified constraint set.

Assumption 1. The pair (A, B) is stabilizable. The sets $\mathcal{W}$ and $\mathcal{Y}$ are compact, convex, and contain the origin in their interiors.

Equilibria of (1) satisfy $Z[\bar{x}^T \ \bar{u}^T \ \bar{z}^T]^T = 0$, where

$$Z = \begin{bmatrix} A - I & B & 0 \\ E & F & -I \end{bmatrix}.$$

Moreover, these equilibria can be parameterized by an auxiliary reference $v \in \mathbb{R}^v$ according to

$$\begin{bmatrix} \bar{x}_v \\ \bar{u}_v \\ \bar{z}_v \end{bmatrix} = \begin{bmatrix} G_x \\ G_u \\ G_z \end{bmatrix} v, \quad (4)$$

where $G^T = [G_x^T \ G_u^T \ G_z^T]^T$ is a basis for the nullspace of Z and **Assumption 1** ensures that $\text{Null}(Z) \neq \{0\}$ [25]. The following assumption excludes ill-posed reference tracking problems.

Assumption 2. The matrix G_z is full rank.

As discussed in [22], the use of an auxiliary reference v allows for handling of over/under parameterization of the equilibrium manifold. We define the set of strictly steady-state auxiliary references and target setpoints as

$$\begin{aligned} \mathcal{V}_\epsilon &:= \{v \in \mathbb{R}^v \mid G_y v \in (1 - \epsilon)\mathcal{Y}\}, \\ \mathcal{R}_\epsilon &:= \{G_z v \in \mathbb{R}^z \mid v \in \mathcal{V}_\epsilon\}, \end{aligned}$$

where $G_y = CG_x + DG_u$ and $0 < \epsilon \ll 1$ is a fixed parameter. The set of strictly steady-state admissible equilibria is

$$\Sigma := \{(x, v) \mid x = G_x v, v \in \mathcal{V}_\epsilon\}. \quad (5)$$

The following optimal control problem defines the PMPC feedback law used to accomplish the stated control objective:

$$\min_{\hat{\mathbf{u}}} J_N(x, \hat{\mathbf{w}}, \hat{\mathbf{u}}, v) = V_f(\hat{x}_N, v) + \sum_{i=0}^{N-1} \ell(\hat{x}_i, \hat{u}_i, \hat{w}_i, v) \quad (6a)$$

$$\text{s.t. } \hat{x}_0 = x, \quad (6b)$$

$$\hat{x}_{i+1} = A\hat{x}_i + B\hat{u}_i + \hat{w}_i, \quad i \in \mathbb{N}_{[0, N-1]}, \quad (6c)$$

$$C\hat{x}_i + D\hat{u}_i \in \mathcal{Y}, \quad i \in \mathbb{N}_{[0, N-1]}, \quad (6d)$$

$$(\hat{x}_N, v) \in \mathcal{T}. \quad (6e)$$

The control sequence $\hat{\mathbf{u}} = (\hat{u}_0, \dots, \hat{u}_{N-1})$ is the optimization vector of (6), the predicted state sequence $\hat{\mathbf{x}} = (\hat{x}_0, \dots, \hat{x}_N)$ is parameterized by $\hat{\mathbf{u}}$, and the disturbance sequence $\hat{\mathbf{w}} = (\hat{w}_0, \dots, \hat{w}_{N-1})$ is an input parameter. We use the notation $\mathcal{P}_N(x, \hat{\mathbf{w}}, v)$ to refer to problem (6) specified for an initial state x , a disturbance sequence $\hat{\mathbf{w}}$, and a reference v . The stage and terminal costs are

$$\ell(x, u, w, v) = \|x - G_x v\|_Q^2 + \|u - G_u v\|_R^2 + \|w\|_S^2,$$

$$V_f(x, v) = \|x - G_x v\|_P^2,$$

where G_x and G_u are defined in (4). The inclusion of w in the stage cost has no effect on the solutions to (6). However, the inclusion of this term allows for the optimal cost of $\mathcal{P}_N(x, \hat{\mathbf{w}}, v)$ to be bounded by a class $\mathcal{K}_\infty$ function of $\|(x - G_x v, \hat{\mathbf{w}})\|$, which facilitates the ISS analysis in Section 3.

Assumption 3. The cost matrices satisfy $Q \geq 0$ with (A, Q) detectable, $R > 0$, and $S > 0$.

The feasible set $\mathcal{F}(x, \hat{w}, v)$, the set of feasible parameters Γ_N , and the value function $V_N : \Gamma_N \rightarrow [0, \infty)$ of $\mathcal{P}_N(x, \hat{w}, v)$ are defined as

$$\mathcal{F}(x, \hat{w}, v) = \{\hat{u} \in \mathbb{R}^{n_u N} \mid \text{6b--6e hold}\}, \quad (7)$$

$$\Gamma_N = \{(x, \hat{w}, v) \mid \mathcal{F}(x, \hat{w}, v) \neq \emptyset\}, \quad (8)$$

$$V_N(x, \hat{w}, v) = \min_{\hat{u} \in \mathcal{F}(x, \hat{w}, v)} J_N(x, \hat{w}, \hat{u}, v), \quad (9)$$

where we also define the subset

$$\Lambda = \Gamma_N \cap (\mathbb{R}^n \times \mathcal{W}^N \times \mathcal{V}_e). \quad (10)$$

For an arbitrary sequence of reference commands $\{v_k\}$, the closed-loop system under PMPC feedback is

$$x_{k+1} = Ax_k + Bu_k + w_k, \quad (11a)$$

$$u_k = u^*(x_k, \mathbf{w}_k, v_k), \quad (11b)$$

$$\mathbf{w}_k = (w_k, \dots, w_{k+N-1}), \quad (11c)$$

where $u^*(x_k, \mathbf{w}_k, v_k) = \hat{u}_{0|k}^* = \hat{u}_0^*(x_k, \mathbf{w}_k, v_k)$ denotes³ the first control input given by optimal solution to $\mathcal{P}_N(x_k, \mathbf{w}_k, v_k)$.

Remark 1. We adopt identical assumptions to [20] regarding the availability of disturbance preview information. Specifically, we assume that an exact preview of the disturbance is available over the horizon of the PMPC problem, while no information is available outside of the horizon. As discussed in [20, Remark 1], this is reasonable a representation in applications where information is available from sensors such as RADAR, LIDAR, and cameras. While this assumption may be restrictive in other cases, the primary goal of this paper is to motivate the combination of the FG and PMPC. Hence, we adopt a problem setting where the results are most easily interpreted and claim that the provided insights will still hold for other PMPC formulations (e.g., [21]). Moreover, the FG is employed with the intention of making this assumption more palatable by allowing for the use of shorter prediction horizons; thereby reducing the duration for which preview information is required.

In this framework, [20] showed that the closed-loop system under regulatory PMPC is ISS if $(x_0, \mathbf{w}_0, 0) \in \Gamma_N$. As discussed in the introduction, enforcing that $(x_0, \mathbf{w}_0, 0) \in \Gamma_N$ holds may require the use of a long horizon N . In what follows, we show that recursive feasibility of the PMPC problem can be enforced using smaller horizon lengths by using a FG. To facilitate this development, we first establish nominal ISS properties of (11) for a constant reference $v_k \equiv v$.

3. ISS of tracking MPC with preview information

In this section, we modify the result of [20] to fit the tracking formulation in (6). To begin, we assume that the terminal set is robustly control invariant.

Assumption 4. For all $v \in \mathcal{V}_e$, the terminal set $\mathcal{T}$ satisfies $S_x(\Sigma, v) \subset \text{Int } S_x(\mathcal{T}, v)$, where Σ is defined in (5). Moreover, there exists $u_f : \mathcal{T} \rightarrow \mathbb{R}^{n_u}$ and $\sigma_1 \in \mathcal{K}_\infty$ such that for all $(x, v) \in \mathcal{T}$:

$$(Ax + Bu_f(x, v) + w, v) \in \mathcal{T}, \quad \forall w \in \mathcal{W},$$

$$Cx + Du_f(x, v) \in \mathcal{V},$$

$$V_f(Ax + Bu_f(x, v), v) - V_f(x, v) \leq -\ell(x, u_f(x, v), 0, v) - \sigma_1(\|x - G_x v\|). \quad (12)$$

³ We use the notation $(\hat{x}_{0|k}^*, \dots, \hat{x}_{N|k}^*)$, $(\hat{u}_{0|k}^*, \dots, \hat{u}_{N-1|k}^*)$, and $(\hat{w}_{0|k}, \dots, \hat{w}_{N-1|k})$ to refer to the optimal trajectories and the disturbance preview of $\mathcal{P}_N(x_k, \mathbf{w}_k, v_k)$. Note that $\hat{w}_{i|k} = w_{k+i}$ in this context. When necessary, we also use the notation $\hat{x}_i^*(x, \mathbf{w}, v)$ and $\hat{u}_i^*(x, \mathbf{w}, v)$ to refer to the i^{th} predicted state and control resulting from the optimal solution to $\mathcal{P}_N(x, \mathbf{w}, v)$.

Remark 2. One feasible choice of terminal ingredients is:

- $u_f(x, v) = G_u v - K(x - G_x v)$, where $A - BK$ is a Schur matrix,
- $\mathcal{T}$ is the maximal disturbance invariant set [26] for the $n + n_v$ dimensional system $x^+ = Ax + Bu_f(x, v) + w$ and $v^+ = v$ subject to (2) and (3),
- $P = (A - BK)^T P (A - BK) + \lambda_1(Q + K^T R K) + \lambda_2 \Delta$ for some $\Delta > 0$, $\lambda_1 > 1$, and $\lambda_2 > 1$.

Then, (12) holds with $\sigma_1(\|x\|) = \|x\|_A^2$ [20]. Moreover, $\mathcal{T}$ is a compact polyhedron if Assumption 1 holds and $\mathcal{W}$ and $\mathcal{V}$ are polyhedra. The pair $(K, \mathcal{T})$ could also be obtained by optimizing K to maximize the volume of $\mathcal{T}$ (see e.g., [27]).

Proposition 1. The terminal dynamics $x^+ = Ax + Bu_f(x, v) + w$ are ISS with respect to the disturbance w . That is, $\exists \xi_1 \in \mathcal{K}_\infty$ such that $\forall (x, v) \in \mathcal{T}$:

$$V_f(x^+, v) - V_f(x, v) \leq -\ell(x, u_f(x, v), 0, v) - \sigma_1(\|x - G_x v\|) + \xi_1(\|w\|).$$

Proof. Let $(x, v) \in \mathcal{T}$, $x^* = Ax + Bu_f(x, v)$, and note that $(x^*, v) \in \mathcal{T}$ and $(x^+, v) \in \mathcal{T}$ by Assumption 4. Recall that $V_f(x, v) = \|x - G_x v\|_P^2$, and so $V_f(x^+, v) \leq V_f(x^*, v) + 2\|x^* - G_x v\|_P \|w\|_P + \|w\|_P^2$.

Since $\mathcal{T}$ is bounded, $\exists \delta_1 > 0$ such that $\|x - G_x v\|_P \leq \delta_1$ for all $(x, v) \in \mathcal{T}$. Hence, the result follows by defining $\xi_1(\|w\|) = 2\delta_1 \|w\|_P + \|w\|_P^2$ and using (12) to bound $V_f(x^*, v)$. $\square$

Lemma 1. Let $(x_k, \mathbf{w}_k, v_k) \in \Lambda$, where Λ is defined in (10), and let x_{k+1} be given by (11). Then, $(x_{k+1}, \mathbf{w}_{k+1}, v_k) \in \Lambda$.

Proof. The proof is omitted for brevity since it is analogous to that of Theorem 2 in Section 4 or [20, Theorem 1]. $\square$

Theorem 1. Define $\mathbf{z}_k := (x_k, \mathbf{w}_k)$,⁴ $\mathbf{G}_z := [G_x \ 0]^T$, and $\mathbf{z}_v := \mathbf{G}_z v$. Let $(\mathbf{z}_0, v) \in \Lambda$ and consider the closed-loop system in (11) with $v_k \equiv v$. Then, $(\mathbf{z}_k, v) \in \Lambda$ for all $k \geq 0$ and the closed-loop system is ISS with respect to the disturbance signal. More precisely, $\exists c_1, c_3 \in \mathcal{K}_\infty$ such that

$$c_1(\|\mathbf{z}_k - \mathbf{z}_v\|) \leq V_N(\mathbf{z}_k, v) \leq c_3(\|\mathbf{z}_k - \mathbf{z}_v\|), \quad (13)$$

and $\exists \rho \in \mathcal{K}_\infty$ such that

$$V_N(\mathbf{z}_{k+1}, v) - V_N(\mathbf{z}_k, v) \leq -c_1(\|\mathbf{z}_k - \mathbf{z}_v\|) - \sigma_1(\|\hat{x}_{N|k}^* - \bar{x}_v\|) + \rho(\|\mathbf{w}_{k+1}\|). \quad (14)$$

Proof. The result was proven in [20, Theorem 2] for the case of $v = 0$. Hence, we proceed by replicating the proof of [20, Theorem 2] for the general case of $v \in \mathbb{R}^{n_v}$. Using identical steps as in [20, Theorem 2], one can show that (13) holds with

$$c_1(\|\mathbf{z}_k - \bar{\mathbf{z}}_v\|) := \|x_k - \bar{x}_v\|_Q^2 + \|\hat{u}_{0|k}^* - \bar{u}_v\|_R^2 + \sum_{i=0}^{N-1} \|\hat{w}_{i|k}\|_S^2, \quad (15a)$$

$$c_3(\|\mathbf{z}_k - \bar{\mathbf{z}}_v\|) := c_2(\|\mathbf{z}_k - \bar{\mathbf{z}}_v\|) + \xi_2(\|\mathbf{z}_k - \bar{\mathbf{z}}_v\|), \quad (15b)$$

where $c_2, \xi_2 \in \mathcal{K}_\infty$ are functions satisfying $\forall (x, \mathbf{w}, v) \in \Lambda$:

$$|V_N(x, \mathbf{w}, v) - V_N(x, 0, v)| \leq \xi_2(\|\mathbf{w}\|),$$

$$V_N(x, 0, v) \leq c_2(\|x - \bar{x}_v\|).$$

To prove (14), let $(x_k, \mathbf{w}_k, v) \in \Lambda$ and consider the shifted and padded sequence

$$\tilde{\mathbf{u}}_{k+1} = (\hat{u}_{1|k}^*, \dots, \hat{u}_{N|k}^*, u_f(\hat{x}_{N|k}^*, v)),$$

⁴ Henceforth, the arguments of maps (e.g., V_N , J_N , and $\mathcal{F}$) that depend on $(x_k, \mathbf{w}_k)$ will be replaced with $\mathbf{z}_k$. We will revert to the previous notation only when necessary.

which satisfies $\bar{\mathbf{u}}_{k+1} \in \mathcal{F}(\mathbf{z}_{k+1}, v)$ by Lemma 1. Then,

$$\begin{aligned} V_N(\mathbf{z}_{k+1}, v) - V_N(\mathbf{z}_k, v) &\leq J_N(\mathbf{z}_{k+1}, \bar{\mathbf{u}}_{k+1}, v) - V_N(\mathbf{z}_k, v) \\ &= \ell(\hat{x}_{N|k}^*, \hat{u}_{N-1|k+1}^*, \hat{w}_{N-1|k+1}, v) + V_f(\hat{x}_{N|k+1}, v) - V_f(\hat{x}_{N|k}^*, v) \\ &\quad - \ell(x_k, \hat{u}_{0|k}^*, w_k, v) \\ &\leq -\sigma_1(\|\hat{x}_{N|k}^* - \bar{x}_v\|) + \xi(\|\hat{w}_{N-1|k+1}\|) - \ell(x_k, \hat{u}_{0|k}^*, w_k, v), \end{aligned}$$

where $\xi(\|w\|) := \|w\|_S^2 + \xi_1(\|w\|)$ and $\xi_1 \in \mathcal{K}_\infty$ is defined in Proposition 1. The first line follows by feasibility of $\bar{\mathbf{u}}_{k+1}$, the second line by definition of $\bar{\mathbf{u}}_{k+1}$, and the last line by Assumption 4 and Proposition 1. The ISS result in (14) is obtained by defining $\rho(\|\mathbf{w}_{k+1}\|) := \xi(\|\hat{w}_{N-1|k+1}\|) + \sum_{i=0}^{N-2} \|\hat{w}_{i|k+1}\|_S^2$, and employing the substitution

$$\ell(x_k, \hat{u}_{0|k}^*, w_k, v) = c_1(\|\mathbf{z}_k - \bar{\mathbf{z}}_v\|) - \sum_{i=0}^{N-1} \|\hat{w}_{i|k}\|_S^2,$$

which follows by the definition of c_1 in (15a). $\square$

4. Feasibility governor

In this section, we tailor the FG of [22] to supervise the feasibility of $\mathcal{P}_N(x_k, \mathbf{w}_k, v_k)$. We consider the closed-loop system in (11) when the sequence of reference commands $\{v_k\}_{k=1}^\infty$ is specified according to

$$v_{k+1} = g(\hat{x}_N^*(x_k, \mathbf{w}_k, v_k), r), \quad (16)$$

where $v_0 \in \mathcal{V}_e$ is given, and

$$g(\hat{x}_N, r) = \arg \min_{v \in \mathcal{V}_e} \{\psi(v, r) \mid (\hat{x}_N, v) \in \mathcal{T}\}, \quad (17a)$$

$$\psi(v, r) = \begin{cases} \|G_z v - r\|_L^2 & \text{if } G_z \text{ is injective,} \\ \|v - v_r^*\|_L^2 & \text{otherwise,} \end{cases} \quad (17b)$$

$$v_r^* = \arg \min_{v \in \mathcal{V}_e} \|G_z v - r\|_L^2, \quad (17c)$$

where $L > 0$. Note that $G_z v_r^* = r$ if $r \in \mathcal{R}_e$, otherwise $G_z v_r^* \in \arg \min_{s \in \mathcal{R}_e} \|s - r\|_L^2$. Moreover, the function $\psi(\cdot, r)$ is strongly convex [22], and if $\mathcal{T}$ is polyhedral (see Remark 2), then (17a) is an n_v -dimensional quadratic program.

The primary difference in implementation of the FG in (16)–(17) compared to the one in [22] is that $\mathcal{T}$ is an RCI set. Moreover, the closed-loop system resulting from (11) and (16) is quite different from the system considered in [22] due to the presence of the disturbance w_k in the dynamics and the inclusion of the disturbance preview $\mathbf{w}_k$ in the PMPC problem. Hence, we aim to prove that the resulting closed-loop system has the following properties:

- Recursive feasibility, i.e., $(\mathbf{z}_k, v_k) \in \Lambda \implies (\mathbf{z}_{k+1}, v_{k+1}) \in \Lambda$,
- Finite-time convergence of the auxiliary reference command, i.e., $\exists k^* \in \mathbb{N}$ such that $v_k = v_r^*$ for all $k \geq k^*$,
- Input-to-state stability, i.e., the error signal $\|\mathbf{z}_k - \mathbf{G}_z v_r^*\|$ is ISS with respect to $\|\mathbf{w}_k\|$.

First, we demonstrate the recursive feasibility property.

Theorem 2 (Recursive Feasibility). *Consider the closed-loop system (11) with $(\mathbf{z}_k, v_k) \in \Lambda$. Then, a feasible solution to (16) exists and $(\mathbf{z}_{k+1}, v_{k+1}) \in \Lambda$ for v_{k+1} given by (16).*

Proof. Fix $k \in \mathbb{N}$ and let $(\mathbf{z}_k, v_k) \in \Lambda$. Then, $(\hat{x}_{N|k}^*, v_k) \in \mathcal{T}$ by definition of $\mathcal{P}_N(\mathbf{z}_k, v_k)$. Hence, v_k is a feasible solution to (16). So, let $v_{k+1} = g(\hat{x}_{N|k}^*, v_k)$ and define the candidate solution sequence

$$\begin{aligned} \bar{\mathbf{u}}_{k+1} &= (\bar{u}_{0|k+1}, \dots, \bar{u}_{N-2|k+1}, \bar{u}_{N-1|k+1}) \\ &:= (\hat{u}_{1|k}^*, \dots, \hat{u}_{N|k}^*, u_f(\hat{x}_{N|k}^*, v_{k+1})), \end{aligned}$$

and note that the resulting state sequence is

$$\bar{\mathbf{x}}_{k+1} = (\bar{x}_{0|k+1}, \dots, \bar{x}_{N-1|k+1}, \bar{x}_{N|k+1}) := (\hat{x}_{1|k}^*, \dots, \hat{x}_{N|k}^*, \bar{x}_{N|k+1}),$$

where $\bar{x}_{N|k+1} = A\hat{x}_{N|k}^* + Bu_f(\hat{x}_{N|k}^*, v_{k+1}) + \hat{w}_{N-1|k+1}$. Observe that $C\bar{x}_{i|k+1} + D\bar{u}_{i|k+1} = C\hat{x}_{i+1|k}^* + D\hat{u}_{i+1|k}^* \in \mathcal{Y}$ for all $i \in \mathbb{N}_{[0, N-2]}$ since $(\mathbf{z}_k, v_k) \in \Lambda$. Moreover, $(\bar{x}_{N-1|k+1}, v_{k+1}) = (\hat{x}_{N|k}^*, v_{k+1}) \in \mathcal{T}$ by definition of v_{k+1} . Therefore, $C\bar{x}_{N-1|k+1} + D\bar{u}_{N-1|k+1} = C\bar{x}_{N-1|k+1} + Du_f(\bar{x}_{N-1|k+1}, v_{k+1}) \in \mathcal{Y}$ and $(\bar{x}_{N|k+1}, v_{k+1}) \in \mathcal{T}$ by Assumption 4. Thus, $\bar{\mathbf{u}}_{k+1} \in \mathcal{F}(\mathbf{z}_{k+1}, v_{k+1})$ and $(\mathbf{z}_{k+1}, v_{k+1}) \in \Lambda$. $\square$

Next, we establish that the system is ISS with respect to the combined effect of the disturbance and reference change.

Lemma 2. *Consider the closed-loop system in (11) with $(\mathbf{z}_k, v_k) \in \Lambda$. Then, $\exists \sigma_v \in \mathcal{K}_\infty$ such that for all $v_{k+1} \in \mathcal{V}_e$:*

$$\begin{aligned} V_N(\mathbf{z}_{k+1}, v_{k+1}) - V_N(\mathbf{z}_k, v_k) &\leq V_N(\mathbf{z}_{k+1}, v_k) - V_N(\mathbf{z}_k, v_k) + \sigma_v(\|\Delta v_k\|) \\ &\leq -c_1(\|\mathbf{z}_k - \mathbf{G}_z v_k\|) - \sigma_1(\|\hat{x}_{N|k}^* - G_x v_k\|) + \rho(\|\mathbf{w}_{k+1}\|) + \sigma_v(\|\Delta v_k\|), \end{aligned}$$

where $\Delta v_k := v_{k+1} - v_k$.

Proof. Under the established assumptions, V_N is uniformly continuous [6, Proposition 1]. Thus, by definition of uniform continuity, $\exists \sigma_v \in \mathcal{K}_\infty$ such that $|V_N(\mathbf{z}, v') - V_N(\mathbf{z}, v)| \leq \sigma_v(\|v' - v\|)$ for all $v \in \mathcal{V}_e$. Hence,

$$\begin{aligned} V_N(\mathbf{z}_{k+1}, v_{k+1}) - V_N(\mathbf{z}_k, v_k) &= V_N(\mathbf{z}_{k+1}, v_k) - V_N(\mathbf{z}_k, v_k) + V_N(\mathbf{z}_{k+1}, v_{k+1}) - V_N(\mathbf{z}_{k+1}, v_k) \\ &\leq V_N(\mathbf{z}_{k+1}, v_k) - V_N(\mathbf{z}_k, v_k) + \sigma_v(\|\Delta v_k\|), \end{aligned}$$

where $V_N(\mathbf{z}_{k+1}, v_k) - V_N(\mathbf{z}_k, v_k)$ is bounded by (14). $\square$

The following Lemma from [22] states that the reference step is lower-bounded whenever $(\hat{x}_{N|k}^*, v_k)$ is inside a region in the interior of $\mathcal{T}$.

Lemma 3 ([22, Lemma 1]). *Consider the system (11) and let*

$$S_\delta(\Sigma) = \{(x, v) \mid v \in \mathcal{V}_e, V_f(x, v) \leq \delta\},$$

where $S_0(\Sigma) := \Sigma$. Then, given $\delta > 0$ satisfying $S_\delta(\Sigma) \subset \text{Int } \mathcal{T}$, $\exists \alpha > 0$ such that if $(\hat{x}_{N|k}^*, v_k) \in S_\delta(\Sigma)$ and v_{k+1} is given by (16), then

$$\begin{cases} \|v_{k+1} - v_k\| \geq \alpha & \text{if } \|v_k - v_r^*\| > \alpha, \\ v_{k+1} = v_r^* & \text{if } \|v_k - v_r^*\| \leq \alpha. \end{cases}$$

Proof. The result follows directly from [22, Lemma 1] since the terminal set $\mathcal{T}$ satisfies [22, Assumption 5] by Assumption 4. $\square$

Next, we establish conditions on the disturbance that guarantee $(\hat{x}_{N|k}^*, v_k) \in S_\delta(\Sigma)$ can eventually be satisfied. Define a fixed constant $\delta > 0$ satisfying $S_\delta(\Sigma) \subset \text{Int } \mathcal{T}$, and define $d_1 > 0$ to be a constant satisfying

$$\sigma_1(\|x - \bar{x}_v\|) \geq d_1, \quad \forall (x, v) \notin S_\delta(\Sigma), \quad (18)$$

where σ_1 is defined in Assumption 4. The following Lemma states if $(\hat{x}_{N|k}^*, v_k) \notin S_\delta(\Sigma)$ and $\hat{w}_{N|k}$ is sufficiently small, then the optimal cost is strictly decreasing.

Lemma 4. *Let the stated assumptions hold, $(\mathbf{z}_k, v) \in \Lambda$, $v_{k+1} = v_k = v$, and suppose that $(\hat{x}_{N|k}^*, v) \notin S_\delta(\Sigma)$ while $\xi(\|\hat{w}_{N|k}\|) \leq \bar{w}$ for some $\bar{w} \in (0, d_1)$. Then, $\exists c_4 \in \mathcal{K}_\infty$ such that*

$$V_N(\mathbf{z}_{k+1}, v) - V_N(\mathbf{z}_k, v) \leq -c_4(\|\mathbf{z}_k - \mathbf{G}_z v\|). \quad (19)$$

Proof. To begin, define $d_2 = \sup_{w \in \mathcal{W}} \|w\|_S^2$ and let $d_3 \in (0, 1)$ be a constant satisfying $(N-1)d_2 d_3 \leq d_1 - \bar{w}$. By Theorem 1 and the definition of d_1 :

$$V_N(\mathbf{z}_{k+1}, v) - V_N(\mathbf{z}_k, v) \leq -c_1(\|\mathbf{z}_k - \mathbf{G}_z v\|) - d_1 + \rho(\|\mathbf{w}_{k+1}\|), \quad (20)$$

since $(\hat{x}_{N|k}^*, v) \notin S_\delta(\Sigma)$ by assumption. Recall that $\rho(\|\mathbf{w}_{k+1}\|) = \xi(\|\hat{w}_{N-1|k+1}\|) + \sum_{i=0}^{N-2} \|\hat{w}_{i|k+1}\|_S^2$. Let $\rho_1(\mathbf{w}_{k+1}) := \sum_{i=0}^{N-2} \|\hat{w}_{i|k+1}\|_S^2$. Then,

$$\begin{aligned} \rho(\|\mathbf{w}_{k+1}\|) - d_1 &\leq \rho_1(\mathbf{w}_{k+1}) - (d_1 - \bar{w}) \\ &\leq \rho_1(\mathbf{w}_{k+1}) - (N-1)d_2d_3 \\ &\leq (1-d_3)\rho_1(\mathbf{w}_{k+1}) \end{aligned}$$

where the first line follows from $\xi(\|\hat{w}_{N|k}\|) \leq \bar{w}$, the second line follows from the definition of d_3 , and the last line follows from $\|\hat{w}_{i|k+1}\|_S^2 \leq d_2$ for all $i \in \mathbb{N}_{[0, N-1]}$ by definition of d_2 . Combining the bound for $\rho(\|\mathbf{w}_{k+1}\|) - d_1$ and the definition of c_1 in (15a) with (20) yields

$$\begin{aligned} V_N(\mathbf{z}_{k+1}, v) - V_N(\mathbf{z}_k, v) \\ \leq -\ell(x_k, \hat{u}_{0|k}^*, \hat{w}_{0|k}, v) - \sum_{i=1}^{N-1} \|\hat{w}_{i|k}\|_S^2 + (1-d_3) \sum_{j=0}^{N-2} \|\hat{w}_{j|k+1}\|_S^2. \end{aligned}$$

Then, (19) holds with

$$c_4(\|\mathbf{z}_k - \bar{\mathbf{z}}_v\|) := \ell(x_k, \hat{u}_{0|k}^*, \hat{w}_{0|k}, v) + d_3 \sum_{i=1}^{N-1} \|\hat{w}_{i|k}\|_S^2,$$

since $\hat{w}_{i+1|k} = \hat{w}_{i|k+1}$ for all $i \in \mathbb{N}_{[0, N-1]}$. $\square$

Thus, we make the following technical assumption to ensure that the predicted terminal state $\hat{x}_{N|k}^*$ eventually enters the set $S_x(S_\delta(\Sigma), v_k)$.

Assumption 5. For all $k_0 \in \mathbb{N}$ and $\Delta k \in \mathbb{N}$, there exists $k_1 \geq k_0$ such that $\xi(\|w_k\|) \leq \bar{w}$ for all $k \in \mathbb{N}_{[k_1, k_1 + \Delta k]}$, where $\bar{w} \in (0, d_1)$ is a constant and $\xi \in \mathcal{K}_\infty$ is defined in the proof of Theorem 1.

The following corollary provides intuition regarding the consequences of Assumption 5.

Corollary 1. Let the stated assumptions hold, $(\mathbf{z}_{k_0}, v) \in A$, and let $v_k = v$ for all $k \geq k_0$. Then, $\exists k' \geq k_0$ such that $(\hat{x}_{N|k'}^*, v) \in S_\delta(\Sigma)$.

Proof. For the sake of contradiction, assume that $(\hat{x}_{N|k}^*, v) \notin S_\delta(\Sigma)$ for all $k \geq k_0$. The combination of Assumption 5 and Lemma 4 imply that for any $\Delta k \in \mathbb{N}$, $\exists k_1 \geq k_0$ such that $\forall k \in \mathbb{N}_{[k_1, k_1 + \Delta k]}$:

$$V_N(\mathbf{z}_{k+1}, v) - V_N(\mathbf{z}_k, v) \leq -c_4(\|\mathbf{z}_k - \mathbf{G}_z v\|),$$

where $c_4 \in \mathcal{K}_\infty$ is defined in the proof of Lemma 4. Hence, there exists a sufficiently large $\Delta k \in \mathbb{N}$, $k_1 \geq k_0$ and $k_2 \in \mathbb{N}_{[k_1, k_1 + \Delta k]}$ such that $V_N(\mathbf{z}_{k_2}, v) \leq \delta$. Then, $V_f(\hat{x}_{N|k_2}^*, v) \leq V_N(\mathbf{z}_{k_2}, v) \leq \delta$, where the first inequality follows by positive definiteness of the terms in V_N . Thus, $(\hat{x}_{N|k_2}^*, v) \in S_\delta(\Sigma)$ and we have arrived at a contradiction. $\square$

Assumption 5 also ensures that the FG eventually finds a non-trivial update of the reference command.

Lemma 5. Let the stated assumptions hold and let $k_0 \in \mathbb{N}$. Consider the closed-loop system (11) with v_k defined by (16) for all $k > k_0$. If $(\hat{x}_{N|k_0}^*, v_{k_0}) \in \mathcal{T}$, then there exists a constant $\eta > 0$ and a finite time $k_i \geq k_0$ such that

$$\|v_{k_i+1} - v_{k_i}\| \geq \eta \quad \text{or} \quad v_{k_i+1} = v_r^*. \quad (21)$$

Proof. To begin, let $(\hat{x}_{N|k_0}^*, v_{k_0}) \in \mathcal{T}$ and define $\eta \in (0, \min\{\alpha, c_6^{-1}(\delta/2)\})$, where α and δ are defined in Lemma 3 and $c_6 := c_3 \circ c_4^{-1} \circ \sigma_v$. For the sake of contradiction, assume $\nexists k_i \geq k_0$ satisfying (21). By Lemma 3 and the choice of $\eta < \alpha$, it must hold that $(\hat{x}_{N|k}^*, v_k) \notin S_\delta(\Sigma)$ for all $k \geq k_0$. Then, the combination of Assumption 5 and Lemma 4 implies that for any $\Delta k \in \mathbb{N}$, $\exists k_1 \geq k_0$ such that $\forall k \in \mathbb{N}_{[k_1, k_1 + \Delta k]}$:

$$V_N(\mathbf{z}_{k+1}, v_k) - V_N(\mathbf{z}_k, v_k) \leq -c_4(\|\mathbf{z}_k - \mathbf{G}_z v_k\|).$$

Moreover, when combined with Lemma 2, this implies that for all $\Delta k \in \mathbb{N}$, $\exists k_1 \geq k_0$ such that $\forall k \in \mathbb{N}_{[k_1, k_1 + \Delta k]}$:

$$V_N(\mathbf{z}_{k+1}, v_{k+1}) - V_N(\mathbf{z}_k, v_k) < -c_4(\|\mathbf{z}_k - \mathbf{G}_z v_k\|) + \sigma_v(\eta),$$

where we have used the assumption that $\|\Delta v_k\| < \eta$ for all $k \geq k_0$. Thus, there exists a sufficiently large $\Delta k \in \mathbb{N}$, $k_1 \geq k_0$, and $k_2 \in \mathbb{N}_{[k_1, k_1 + \Delta k]}$ such that

$$c_4(\|\mathbf{z}_{k_2} - \mathbf{G}_z v_{k_2}\|) \leq \sigma_v(\eta) + c_5^{-1}(\delta/2), \quad (22)$$

where $c_5 := c_3 \circ c_4^{-1}$. Then,

$$\begin{aligned} V_f(\hat{x}_{N|k_2}^*, v_{k_2}) &\leq V_N(\mathbf{z}_{k_2}, v_{k_2}) \leq c_3(\|\mathbf{z}_{k_2} - \mathbf{G}_z v_{k_2}\|) \\ &\leq c_3 \circ c_4^{-1}(\sigma_v(\eta) + c_5^{-1}(\delta/2)) \\ &= c_6(\eta) + \delta/2 < \delta. \end{aligned}$$

where the first inequality follows by positive definiteness of the terms in V_N , the second by (13), the third by (22), and the final line follows by definition of c_5 , c_6 , and $\eta < c_6^{-1}(\delta/2)$. Therefore, $(\hat{x}_{N|k_2}^*, v_{k_2}) \in S_\delta(\Sigma)$, so we have arrived at a contradiction. $\square$

Remark 3. The intuition provided by Corollary 1 and Lemma 5 is that the effect of the out-of-preview disturbances can be mitigated by selecting u_f and σ_1 (see Remark 2) with the objective of maximizing d_1 in (18). In other words, the terminal ingredients should be selected so that the volume of $S_\delta(\Sigma)$ is maximized and the function σ_1 is “large” in some sense (e.g., if $\sigma_1(\cdot) = \|\cdot\|_2^2$, then we desire $\lambda_{\min}(\Delta) \gg 0$).

Remark 4. Assumption 5 is an important tool used to prove Lemma 5. We make the following notes regarding Assumption 5:

- It is a less restrictive version of the *origin-persistence* condition in [28].
- It is satisfied if, for example, the disturbance elements w_k are statistically independent random variables, w_k is bounded for all $k \in \mathbb{N}$, and the probability that $\xi(\|w_k\|) \leq \bar{w}$ is greater than a positive constant for all $k \in \mathbb{N}$ [28].
- It is stricter than necessary. Specifically, it only needs to hold for all $\Delta k \leq \Delta k_{\max}$ for some $\Delta k_{\max} < \infty$ since A is compact. We do not explicitly prove this since doing so would obfuscate the core results. Moreover, it only needs to hold for all $k_0 \in \mathbb{N}$ such that $v_{k_0} \neq v_r^*$.
- A less restrictive assumption would be to simply assume that $\forall k \in \mathbb{N}$, $\exists k' \geq k$ such that $(\hat{x}_{N|k'}^*, v_{k'}) \in S_\delta(\Sigma)$. However, it is difficult to intuitively understand when such an assumption is satisfied. We use Assumption 5 and Remark 3 to provide intuition behind how the disturbance effects the convergence of v_k .
- Assumption 4 implicitly limits the size of the disturbance set $\mathcal{W}$. Hence, the properties enforced by Assumption 5 might hold for a given system without any additional restrictions on the disturbance.

The following lemma, whose proof can be found in [22, Lemma 3], follows from strong convexity of ψ .

Lemma 6 ([22, Lemma 3]). Given the stated assumptions, for all $(\mathbf{z}_k, v_k) \in A$ there exists $h > 0$ such that

$$\|v_{k+1} - v_r^*\|^2 \leq \|v_k - v_r^*\|^2 - h\|v_k - v_{k+1}\|^2, \quad (23)$$

where $v_{k+1} = g(\hat{x}_{N|k}^*, r)$.

Lemma 7 ([22, Lemma 4]). Let the stated assumptions hold and define $M = \lceil \frac{\|v_0 - v_r^*\|}{h\eta^2} \rceil$. If there exists a sequence of M time instances $k_1, \dots, k_M$ such that (21) is satisfied at each k_i , then $v_k = v_r^*$ for all $k > k_M$.

Proof. Let (21) be satisfied for all $k_1, \dots, k_M$. If $g(\hat{x}_{N|k_i}^*, r) = v_r^*$ for any $j < M$, then $g(\hat{x}_{N|k_i}^*, r) = v_r^*$ for all $i \geq j$. Thus, consider the scenario where $\|g(\hat{x}_{N|k_i}^*, r) - v_{k_i}\| \geq \eta$ for all $i \in \mathbb{N}_{[1, M]}$. Then, Lemma 6 implies that

$$\|g(\hat{x}_{N|k_i}^*, r) - v_r^*\|^2 \leq \|v_0 - v_r^*\|^2 - Mh\eta^2 \leq 0.$$

Hence, $\|v_{k_M+1} - v_r^*\|^2 = 0$. $\square$

The following theorems show that $v_k \rightarrow v_r^*$ in finite-time and that the error signal $\|z_k - G_z v_r^*\|$ is ISS.

Theorem 3 (Finite-time Convergence of v_k). *Let the stated assumptions hold and consider the closed-loop system (11) with v_k defined by (16) for all $k > 0$. If $(\hat{x}_{N|0}^*, v_0) \in \mathcal{T}$, then $\exists k^* \geq 0$ such that $v_k = v_r^*$ for all $k \geq k^*$.*

Proof. The proof follows by noting that there always exists a subsequence $\{k_i\}_{i=1}^M$ satisfying Lemma 7 by Lemma 5. $\square$

Theorem 4 (Input-to-state Stability). *Let the stated assumptions hold and consider the closed-loop system (11) with v_k defined by (16) for all $k > 0$. If $(\hat{x}_{N|0}^*, v_0) \in \mathcal{T}$, then the error $\|z_k - G_z v_r^*\|$ is ISS with respect to the disturbances w_k . Moreover, $(z_k, v_k) = (z_k, v_r^*) \in \Lambda$ for all $k \geq k^*$.*

Proof. Given that $\Delta v_k = 0$ for all $k \geq k^*$, then Lemma 2 implies that $\forall k > k^*$:

$$\begin{aligned} V_N(z_{k+1}, v_{k+1}) - V_N(z_k, v_k) &= V_N(z_{k+1}, v_r^*) - V_N(z_k, v_r^*) \\ &\leq -c_1(\|z_k - G_z v_r^*\|) - \sigma_1(\|\hat{x}_{N|k}^* - G_x v_r^*\|) + \rho(\|w_{k+1}\|). \end{aligned}$$

Hence, there exists an asymptotic gain $\gamma \in \mathcal{K}_\infty$ such that [24]:

$$\limsup_{k \rightarrow \infty} \|z_k - G_z v_r^*\| \leq \gamma \left(\limsup_{k \rightarrow \infty} \|w_k\| \right).$$

Moreover, we note that the system is *uniformly bounded input bounded state* (UBIBS) [24] since $(z_k, v_k) \in \Lambda$ for all $k \geq 0$ if $(z_0, v_0) \in \Lambda$. Therefore, there exists an ISS Lyapunov function for the signal $\|z_k - G_z v_r^*\|$ by [24, Theorem 1]. $\square$

Remark 5. Stabilizing the system in (11) without the FG (i.e., executing the MPC law with $v_k \equiv v_r^*$) would require satisfaction of $(z_0, v_r^*) \in \Gamma_N$. The FG relaxes this requirement to $(\hat{x}_{N|0}^*, v_0) \in \mathcal{T}$ (i.e., $(z_0, v_0) \in \Gamma_N$), where $v_0 \in \mathcal{V}_e$ is a free variable. Thus, the FG-PMPC strategy is feasible if the initial state is sufficiently close to any equilibrium. Moreover, the FG-PMPC strategy is guaranteed to maintain feasibility if the target reference r is changed.

Remark 6. It is possible, in principle, to extend these results to the setting of nonlinear PMPC by extending the nonlinear PMPC analysis in [20] to the nonlinear reference tracking MPC formulation of [29] (while adopting the assumptions of both [20,29]). Under such a formulation, Assumption 5 would be modified to enforce a similar restriction on the disturbance sequence $\{w_k\}$ for some function $\xi \in \mathcal{K}_\infty$ that follows from assumed continuity properties of the nonlinear dynamics (i.e., the function δ defined in the proof of [20, Theorem 2]). However, computation of an RCI terminal set $\mathcal{T}$ satisfying Assumption 4 for a nonlinear dynamic system is a significant practical challenge. Hence, formulating the FG optimization problem in (17) would be much more difficult. We explore the specific case of linear systems in this paper since in such a case it is possible to easily compute terminal ingredients satisfying Assumption 4.

5. Example

An adaptive cruise control example is used to demonstrate the combined FG-PMPC strategy. The example is motivated by vehicle platooning problems (e.g., [30]). The control task is to maintain a follower vehicle a set distance behind a leader vehicle and to optimize fuel efficiency. We assume the leader communicates its future control intention to the follower, but that the follower has no influence on the leader's actions.

Let (s_l, v_l) and (s, v) represent the position and velocity of the leader and follower relative to some fixed point behind the follower. The control task is to command the follower's acceleration a so that $e_s := s_l - \delta_s - s$ and $e_v := v_l - v$ are near zero, where $\delta_s = 2$ m is the desired

following distance. The continuous-time dynamics can be modeled as a double integrator:

$$\begin{bmatrix} \dot{e}_s \\ \dot{e}_v \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} e_s \\ e_v \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \end{bmatrix} (a - a_l),$$

where a_l is the leader's acceleration command, which acts as the previewed "disturbance" in this example. The system is discretized using a sampling period of $T = 0.04$ sec. Define $x_k = (e_s, e_v)(Tk)$, $u_k = a(Tk)$, and $w_k = a_l(Tk)$. The tracking output is the position error, i.e., $E = [1 \ 0]$ and $F = 0$ in (1), and the desired setpoint is $r = 0$. Note that $G_z = 1$, hence $v_r^* = r = 0$. The initial state is $x_0 = (30, 0)$.

The OCP in (6) uses weight matrices of $Q = \text{diag}(10, 0.1)$ and $R = 100$. The follower's acceleration is constrained to satisfy $u_k \in \mathcal{U} = [-5, 3]$ m/s², which corresponds to limits characteristic of aggressive driving behavior (e.g., see [31, Figure 2]). In other words, aggressive control actions are penalized heavily in the cost, but allowed when necessary. Meanwhile, the leader's acceleration is constrained to satisfy $w_k \in [-2.5, 1.5]$ m/s². In addition, the follower is required to satisfy an overshoot constraint $e_s \geq -\delta_s/2 = -1$ m to ensure that it does not come close to colliding with the leader.

The terminal control law is $u_f(x, v) = G_u v - K(x - G_x v)$, where K the LQR gain matrix resulting from $Q_{\text{LQR}} = I$ and $R_{\text{LQR}} = 0.1$. The terminal set $\mathcal{T}$ is determined by computing the maximal disturbance invariant set [26] for the system:

$$\begin{bmatrix} x_+ \\ v_+ \end{bmatrix} = \begin{bmatrix} A - BK & B(G_u + KG_x) \\ 0 & I \end{bmatrix} \begin{bmatrix} x \\ v \end{bmatrix} - \begin{bmatrix} B \\ 0 \end{bmatrix} w.$$

The terminal cost matrix P of the MPC problem is then defined as the positive definite solution of $P = (A - BK)^T P (A - BK) + \lambda_1(Q + K^T R K) + \lambda_2 \Delta$ for $\Delta = I$ and $\lambda_1 = \lambda_2 = 1.1$. Different weight matrices are used for the MPC and terminal control laws since the maximal disturbance invariant set is empty when K is selected as the LQR gain for Q and R .

Four different MPC feedback policies will be compared:

1. MPC: $u_k = u^*(x_k, 0, v_r^*)$.
2. PMPC: $u_k = u^*(x_k, w_k, v_r^*)$.
3. FG-MPC: $u_k = u^*(x_k, 0, g(\hat{x}_{N|k}^*, v_r^*))$.
4. FG-PMPC: $u_k = u^*(x_k, w_k, g(\hat{x}_{N|k}^*, v_r^*))$.

In other words, we compare MPC without preview (MPC), MPC with preview (PMPC), feasibility governed MPC without preview (FG-MPC), and feasibility governed MPC with preview (FG-PMPC) — the latter being the strategy we propose in this paper. The FG-based policies use an initial reference of $v_0 = 30$. Hence, the FG-PMPC policy is recursively feasible for all $N \geq 1$ in this example since $(x_0, v_0) \in \mathcal{T}$.

5.1. Effect of preview information

The first case study demonstrates the importance of preview information by showing a scenario in which the MPC and FG-MPC policies lead to severe violation of the system constraints.⁵ The disturbance is defined by $w_k = -2.5$ for $kT \in [0, 6]$ sec, $w_k = 1.5$ for $kT \in [14, 22]$ sec, and $w_k = 0$ otherwise (see Fig. 2). More specifically, the leader reduces its speed by 15 m/s during the interval $kT \in [0, 6]$ sec by negatively saturating its acceleration. Then, after a period of constant velocity, the leader increases its speed by 12 m/s by positively saturating its acceleration in the interval $kT \in [14, 22]$ sec. The MPC and PMPC policies use a horizon length of $N = 102$, which is the minimum value necessary to enforce $(x_0, w_0, v_r^*) \in \Gamma_N$ for the specified disturbance sequence. In contrast, the FG-based policies use a shortened horizon of $N = 51$.

⁵ The MPC and FG-MPC simulations are extended by softening the state and terminal constraints at timesteps where $(x_k, 0, v_k) \notin \Gamma_N$. The constraint slacks are penalized using a quadratic penalty term with a weighting of 10^8 .

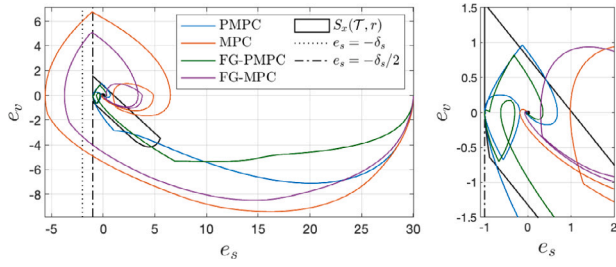


Fig. 1. Case 1: Phase plane trajectories for each feedback policy.

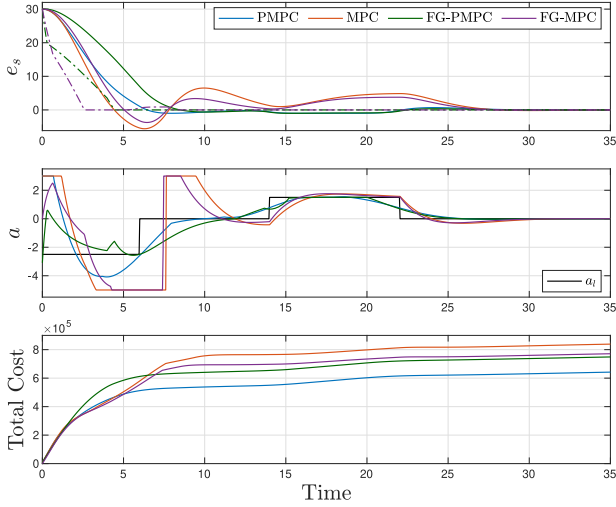


Fig. 2. Case 1: Closed-loop trajectories for each feedback policy. The colored dotted lines in the top plot correspond to the reference trajectories v_k generated by the FG-MPC and FG-PMPC policies. The solid black line in the middle plot is the leader's acceleration a_l . The total cost at time $t_k = Tk$ is defined by $\sum_{i=0}^k \|x_i - G_x v_r^*\|_Q^2 + \|u_i - G_u v_r^*\|_R^2$.

The closed-loop trajectories that result from each feedback policy are shown in Figs. 1 and 2. The MPC and FG-MPC policies result in violation of the overshoot constraint $e_s \geq -\delta_s/2$ due to the deceleration of the leader during the transient phase of the maneuver. In fact, the worst-case constraint violations result in $e_s < -\delta_s$ (i.e., $s > s_l$, so the vehicles would collide) in both cases. In contrast, both the PMPC and FG-PMPC policies remain recursively feasible.

The PMPC and FG-PMPC policies also incur the least cost. The PMPC policy performs the best due to the fact that it necessitates the use of a horizon length of $N \geq 102$. Thus, the PMPC policy can plan further into the future and make use of more preview information. However, the use of such a long horizon may be impractical. This is discussed more in the second case study.

5.2. PMPC vs. FG-PMPC

The second case study compares the PMPC and FG-PMPC policies in an example where the leader's acceleration is persistently varying. The disturbance sequence is defined as follows: First, $w_k = 1.5$ for $kT \in [0, 6.67]$ sec, and $w_k = 0$ for $kT \in (6.67, 12]$ sec. Thereafter, the disturbance is incremented by $w_{k+1} = w_k - 0.08$ until $w_k - 0.08 \leq -2.5$. Then, w_k is held constant at -2.5 for 3 sec. Thereafter, the disturbance is incremented by $w_{k+1} = w_k + 0.08$ until $w_k + 0.08 \geq 1.5$. Then, w_k is held constant at 1.5 for 3 sec. The process is then repeated by incrementing by $w_{k+1} = w_k - 0.08$ and so on (see Fig. 4 for reference). Three FG-PMPC policies with $N \in \{30, 50, 90\}$ are compared to a PMPC policy with $N = 148$, which is the minimum value necessary to enforce $(x_0, w_0, v_r^*) \in \Gamma_N$.

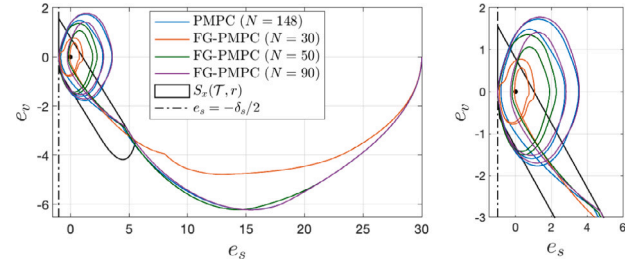


Fig. 3. Case 2: Phase plane trajectories for PMPC ($N = 148$) and several FG-PMPC policies with $N \in \{30, 50, 90\}$.

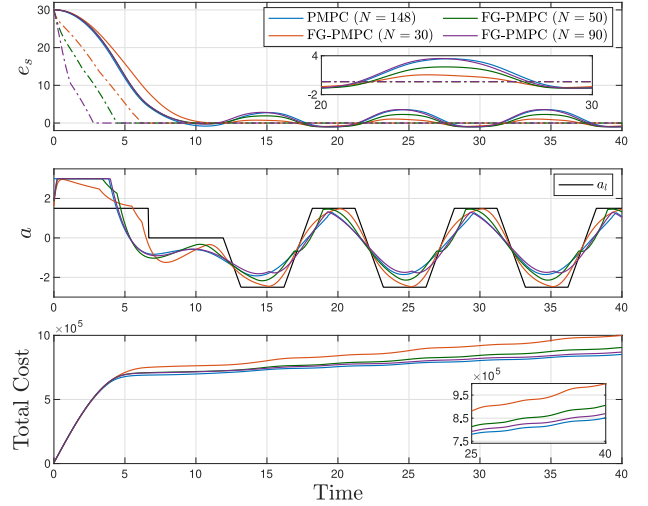


Fig. 4. Case 2: Closed-loop trajectories for PMPC ($N = 148$) and three FG-PMPC policies with $N \in \{30, 60, 90\}$. The colored dotted lines in the top plot correspond to the reference trajectories v_k generated by the FG-PMPC policies. The solid black line in the middle plot is the leader's acceleration a_l .

Figs. 3 and 4 display the closed-loop trajectories that result from each feedback policy. Once again, the PMPC policy incurs the least cost due to the fact that it uses the longest prediction horizon by necessity. In particular, the PMPC policy avoids the use of large control inputs (which are weighted heavily in the cost) by allowing the position error to oscillate between the overshoot constraint and a reasonably close following distance. In contrast, the FG-PMPC policies with $N \in \{30, 50\}$ cannot plan as far ahead or leverage as much preview information as the PMPC policy. As a result, the FG-PMPC policies use slightly larger control inputs both in the transient phase and in steady-state. As the horizon of the FG-PMPC policy is increased, the accumulated cost is reduced and the steady-state trajectories converge to those of the PMPC policy as one would expect (e.g., the $N = 90$ case exhibits similar steady-state performance to the PMPC policy).

We draw attention to the fact that the PMPC policy necessitates the use of a particularly long horizon length of $N \geq 148$ to enforce recursive feasibility in this example. This is caused by leader's positive acceleration during the first 6.67 s of the maneuver. Hence, a long horizon is necessary to enforce that $(x_0, w_0, v_r^*) \in \Gamma_N$ since the disturbance is acting to make reachability to the terminal set more difficult. For comparison, $(x_0, 0, v_r^*) \in \Gamma_N$ is satisfied for all $N \geq 113$ — i.e., the terminal set is more easily reachable when no disturbance is present. As discussed in the introduction, the use of a long prediction horizon poses two critical issues in the setting of PMPC.

Table 1

Mean, median, and 95th percentile execution times (in milliseconds) of each of the control policies in Figs. 3 and 4.

	Mean	Median	95th Perc.
PMPC ($N = 148$)	41.5	32.7	74.5
FG-PMPC ($N = 90$)	18.3	13.4	41.8
FG-PMPC ($N = 50$)	7.17	6.25	13.6
FG-PMPC ($N = 30$)	3.44	3.46	4.13

First, the computational cost of solving the OCP increases with N . To illustrate this point, the execution time⁶ for each control strategy is reported in Table 1. The FG-PMPC policy with $N = 30$ requires ~10% of the computation time of the PMPC policy on average. However, this reduction in computation time comes at the cost of reduced closed-loop performance (e.g., see Figs. 3 and 4). Meanwhile, the FG-PMPC policy with $N = 90$ exhibits very similar closed-loop performance to the PMPC policy, while using less than half the computation time on average. Hence, the horizon length of the FG-PMPC policy can be chosen to trade off closed-loop performance and computational efficiency.

Second, it may be practically infeasible to have preview information over such a long horizon. For reference, a horizon of $N = 148$ with $T = 0.04$ sec means that the leader must communicate its intentions to the follower 5.92 s into the future. Such a requirement could clearly impede implementation of vehicle platooning. In contrast, one is free to select the horizon length of the FG-PMPC policy to be any value smaller or equal to the duration of the time window which preview information is available in practice.

6. Conclusions

This paper extended the feasibility governor (FG) framework of [22] to the preview MPC (PMPC) framework of [20]. It was shown that recursive feasibility of the combined FG-PMPC strategy can be achieved with shorter prediction horizons than a standard PMPC strategy, thereby reducing the amount of computational effort and preview information necessary to implement PMPC. Moreover, conditions were established that guarantee finite-time convergence of the FG reference command and input-to-state stability of the closed-loop system. An adaptive cruise control example was reported to demonstrate a scenario where both preview information and the FG are necessary to ensure safety and real-time feasibility of the PMPC policy.

CRedit authorship contribution statement

Jordan Leung: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. **Ilya Kolmanovsky:** Conceptualization, Resources, Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

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⁶ The simulations were performed using MATLAB R2022b on a 2018 MacBook Pro with a 2.2 GHz 6-Core Intel Core i7 processor and 16 GB 2400 MHz DDR4 RAM. The computation times reported for the FG-PMPC policies include the time to solve both the FG optimization problem in (17) and the resulting PMPC problem. Since $n_v = 1$ in this example, (17) was solved by iterating through the polyhedral constraints in (17a) and constructing the equivalent lower and upper bounds. In each case, the OCPs were solved by expressing (6) in its condensed quadratic form and using MATLAB's quadprog function with default settings. The results were averaged over 50 simulations for consistency.

Data availability

No data was used for the research described in the article.

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